

Protocol for Image processing and analysis of VPS13D recruitment to mitochondria

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Fluorescence images were processed using FIJI (ImageJ; NIH) software. Gaussian blur filters were applied on some of the images presented.

For the analysis of optogenetic experiments:

1. Kymographs were built tracing a line ROI across the mitochondria that was illuminated with Blue Light.
2. An intensity profile was measured by tracing a line ROI across the center of the mitochondrial signal on the kymograph. A background profile was taken using the same line ROI outside the illumination area.
3. The fluorescence intensities were normalized to the min and max values using the following formula: $Normalized\ f[t] = \frac{f[t] - f_{min}[t]}{f_{max}[t] - f_{min}[t]}$, where $f[t] = F_{mito}[t] - F_{bkgd}[t]$.

For the analysis of VPS13D mitochondrial recruitment:

1. Cells coexpressing VPS13D^ΔEGFP and mitoBFP were analyzed individually. Each cell outline was stored as an ROI in the ROI Manger of Fiji
2. “Gaussian Blur” tool and “Enhance Local Contrast” tool were applied to the MitoBFP channel to average the brightness of the whole image and to clarify the shape and margin of mitochondria.
3. An automated macro designed in Fiji was used to measure VPS13D^ΔEGFP signal overlapping with mitochondria and that localized in the cytosol surrounding them.

4. The ratio between the intensity of VPS13D^EGFP on mitochondria and the intensity of VPS13D^EGFP in the cytosol surrounding mitochondria was calculated for each cell and plotted on a graph indicating the enrichment of VPS13D^EGFP on mitochondria.

The macro is included below:

```
macro "ROI in batches auto measure VPS13 binding with mitochondrial " {
```

```
Dialog.create("reminder");//create an interface
```

```
Dialog.addMessage("choose saving path");//an interactive interface for user to set the saving directory
```

```
Dialog.show();//an interactive interface for user to set the saving directory
```

```
dir_saving = getDirectory("Choose a Directory to save");//an interactive interface for user to set the saving directory
```

```
count = roiManager("Count");//save the information from ROI manager into the term "count"
```

```
for (i = 0; i < count; i++){
```

```
    name = getTitle;
```

```
    selectWindow(name);
```

```
    roiManager("Select", i);//save the selected ROIs in the ROI manager in batch
```

```
    Stack.getPosition(channel, slice, frame);//save the selected ROIs in the ROI manager in batch
```

```
    run("Duplicate...", "duplicate slices=" + slice);//save the selected ROIs in the ROI manager in batch
```

```
    saveAs("Tiff", dir_saving + "ROIslice"+i+name); //save the selected ROIs in the ROI manager in batch
```

```
    name = replace(name, ".tif", ""); //motify the name of images
```

```
    C1 = "C1-ROIslice"+i+name+".tif";//motify the name of images
```

```
    C2 = "C2-ROIslice"+i+name+".tif";//motify the name of images
```

```
    run("Split Channels");//split the integrated image into single channel
```

```

Dialog.create("Content of the channel");//create an interface

Dialog.addMessage("Make sure the content of the channels. If same as described, click
OK. If different, click Cancel and make changes.\n");

Dialog.addMessage("Channel 1 is VPS13D channel. Channel 2 is Mitochondrial
channel.");

Dialog.show();//make sure the content of each channels

selectWindow(C1);//choose the cytosol slice

run("Duplicate...", "");//duplicate a seperate image to process

run("Brightness/Contrast...");//preprocess the VPS13D signal in the cytosol area

//measure the intensity of VPS13D signal in mitochondria area

selectWindow(C2);//choose the mitochondria slice

run("Duplicate...", "");//duplicate a seperate image to process

run("Gaussian Blur...", "sigma=2 slice");//balance the brightness of image

run("Enhance Local Contrast (CLAHE)", "blocksize=127 histogram=256 maximum=3
mask=*None* fast_(less_accurate)");//clarify the outline of mitochondria

run("Enhance Local Contrast (CLAHE)", "blocksize=127 histogram=256 maximum=3
mask=*None* fast_(less_accurate)");//clarify the outline of mitochondria

run("Enhance Local Contrast (CLAHE)", "blocksize=127 histogram=256 maximum=3
mask=*None* fast_(less_accurate)");//clarify the outline of mitochondria

run("Threshold...");//choose the mitochondria area

run("Convert to Mask");//generate an accurate mitomask

saveAs("Tiff", dir_saving + "ROI"+i+" mitomask "+name);//generate an accurate mitomask

imageCalculator("AND create", mitomask, C1);//map C1 onto the mitomask

setThreshold(1, 255);//choose the calculated area

saveAs("Tiff", dir_saving + "ROI"+i+" result of mitochondria intensity "+name);

mitointensity=getTitle;

run("Measure");//measure the intensity of VPS13D signal in mitochondria area

//measure the VPS13D intensity in the cytosol area

```

```

run("Dilate");
run("Dilate");
run("Dilate");//duplicate three times to eliminate the interference from mitochondria signal
run("Convert to Mask");
run("Invert");//invert the enlarged mitomask

saveAs("Tiff", dir_saving + "ROI"+i+" mitomask x3 "+name);//generate the enlarged inverted
mitomask

mitomask3 = getTitle;

imageCalculator("AND create", mitomask, mitomask);//map the enlarged inverted mitomask
onto the original mitomask to get the enlarged outline of mitochondria in the cytosol area

saveAs("Tiff", dir_saving + "ROI"+i+" cytomask "+name);//generate outline profile of mitomask,
which refers to mask of cytosol area

cytomask = getTitle;

imageCalculator("AND create", cytomask, C1);//map the cytomask onto cytosol

setThreshold(1,255);

saveAs("Tiff", dir_saving + "ROI"+i+" result of cytosolic intensity "+name);

cytointensity=getTitle;

run("Measure");//measure the VPS13D intensity in the cytosol area

saveAs("Results", dir_saving + "ROI"+i+" measurement results of VPS13 binding to
mitochondrial.csv");

close()
}

Dialog.create("reminder");

Dialog.addMessage("Mission accomplished.Congratulations!");

Dialog.show();

}

```

